

Integration of Network Function Virtualization with the DevOps Pipeline: DevOps Methodology

This seventh instalment of the series on network function virtualization explores the transformative potential of DevOps methodology. Integrating DevOps enables faster SDLC by breaking down silos and fostering a culture of collaboration.



DevOps is seen as a blend of developers (mainly software developers, including programmers, build engineers, testers, and quality engineers) with operators (mainly handling databases, production environment, system administration, and networking). In the book by Ingo M. Weber, Len Bass, and Liming Zhu, 'DevOps: A Software Architect's Perspective', DevOps is defined as: "DevOps is a set of practices intended to reduce the time between committing a change to a system and the change being placed into normal production while ensuring high quality."

The main focus of DevOps is integrating a complete software

development life cycle (SDLC) with the help of automation tools and collaboration between the Dev and Ops teams, enabling them to work collaboratively as a single team and deliver quality software with speed. Adapting the DevOps model of software production helps to push incremental changes and new features to the production servers much faster than traditional methodologies.

Using the right DevOps tools at the right place can help you achieve:

- Accelerated product development
- Faster and more frequent deployments
- Quick incremental releases
- Easy automation
- Seamless collaboration

- Fault tolerance
- Easy monitoring
- Quality assurance
- Fast reaction to feedback

So what is DevOps methodology?

The DevOps methodology can be summed up as narrowing down the gap between the development and operations teams. These teams now come together to produce the deliverables as a single team. The integration of DevOps ensures quick and stable deliverables that align with the organisation's ideology.

Most of these aspects can be achieved through automation of the development cycle. There are many